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CS229 Machine Learning

Linear Models for Regression

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Linear Basis Function Models

- *Regression:* Input vector $\mathbf{x} \equiv (\boldsymbol{x}_1 \cdots \boldsymbol{x}_N)^{\mathrm{T}}$ where $\boldsymbol{x}_n$ is D-dimensional data element and observations $\mathbf{t} \equiv (t_1 \cdots t_N)^{\mathrm{T}}$ where t_n is the corresponding value to $\boldsymbol{x}_n$
- Goal is to predict new value t corresponding to $\boldsymbol{x}$ by modeling predictive distribution $p(t|\boldsymbol{x})$
- Approach: by minimizing the $\mathbb{E}[L]$ of a suitable loss function L
- Simplest regression using linear functions of input variables
- More useful using a linear combination of a fixed set of non-linear functions of the input variables called $\boldsymbol{\phi}(\boldsymbol{x})$
- Models are linear functions of parameters $\boldsymbol{w}$ and non-linear wrt. input variables

Linear Basis Function Models

- *Linear Regression:* linear function of parameters as well as input variables

$$y(\boldsymbol{x}, \boldsymbol{w}) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_D x_D$$

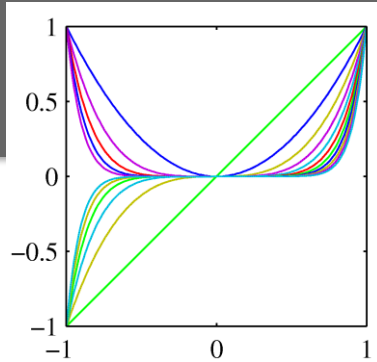
- Extending for considering known basis functions $\phi_j(\boldsymbol{x})$

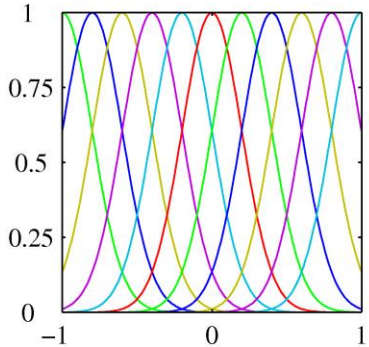
$$y(\boldsymbol{x}, \boldsymbol{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\boldsymbol{x}) = \sum_{j=0}^{M-1} w_j \phi_j(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})$$

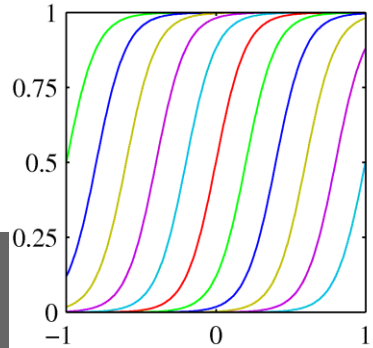
with $\phi_0(\boldsymbol{x}) = 1$, and $\boldsymbol{\phi}(\boldsymbol{x})$ is the feature vector

Linear Basis Function Models

- *Basis Functions:*
- *Polynomial* basis functions (global): $\phi_j(x) = x^j$
- *Gaussian* basis functions (local): $\phi_j(x) = \exp\left\{-\frac{(x-\mu_j)^2}{2s^2}\right\}$ *with locations* μ_j *and scale parameter* s
- *Sigmoidal* basis functions (local): $\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right)$ *where* $\sigma(\mathrm{a}) = \frac{1}{1+\exp(-a)}$ *with locations* μ_j *and slope* s
- *Hyperbolic tangent basis functions:* $\tanh(\mathrm{a}) = 2\sigma(2a) - 1$
- *Fourier basis functions: expansion of sinusoidal functions for modeling frequencies*
- *Wavelet basis functions: modeling levels of detail*







Linear Basis Function Models

- *Maximum Likelihood and Least Squares:*

$$t = y(\boldsymbol{x}, \boldsymbol{w}) + \epsilon \quad p(t|\boldsymbol{x}, \boldsymbol{w}, \beta) = \mathcal{N}(t|y(\boldsymbol{x}, \boldsymbol{w}), \beta^{-1})$$

- For squared loss function the optimal prediction is by conditional mean $\mathbb{E}[t|\boldsymbol{x}] = y(\boldsymbol{x}, \boldsymbol{w})$
- For $\mathbf{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N\}$ $\mathbf{t} = \{t_1, \dots, t_N\}$

$$p(\mathbf{t}|\mathbf{X}, \boldsymbol{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|\boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_n), \beta^{-1})$$

Linear Basis Function Models

- *Maximum Likelihood and Least Squares:*

$$\ln p(\mathbf{t}|\mathbf{X}, \boldsymbol{w}, \beta) = \sum_{n=1}^{N} \ln \mathcal{N}(t_n | \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_n), \beta^{-1})$$

$$= \frac{N}{2} \ln \beta - \frac{N}{2} \ln 2\pi - \beta \frac{1}{2} \sum_{n=1}^{N} \{t_n - \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_n)\}^2$$

- Log likelihood wrt. $\boldsymbol{w}$:

$$\nabla \ln p(\mathbf{t}|\mathbf{X}, \boldsymbol{w}, \beta) = \beta \sum_{n=1}^{N} \{t_n - \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_n)\} \boldsymbol{\phi}(\boldsymbol{x}_n)^T = 0$$

Linear Basis Function Models

- *Maximum Likelihood and Least Squares: normal equations*

$$\sum_{n=1}^{N} t_n \boldsymbol{\phi}(\boldsymbol{x}_n)^T = \boldsymbol{w}^T \sum_{n=1}^{N} \boldsymbol{\phi}(\boldsymbol{x}_n)\boldsymbol{\phi}(\boldsymbol{x}_n)^T \Rightarrow \boldsymbol{w}_{ML} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{t}$$

- Pseudoinverse $\boldsymbol{\Phi}^{\dagger} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T$
- Design Matrix $N \times M$

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\boldsymbol{x}_1) & \cdots & \phi_{M-1}(\boldsymbol{x}_1) \\ \vdots & \ddots & \vdots \\ \phi_0(\boldsymbol{x}_N) & \cdots & \phi_{M-1}(\boldsymbol{x}_N) \end{pmatrix}$$

Linear Basis Function Models

- *Maximum Likelihood and Least Squares: bias term*

$$E_D(\boldsymbol{w}) = \frac{1}{2}\sum_{n=1}^{N}\left\{t_n - w_0 - \sum_{j=1}^{M-1} w_j \phi_j(\boldsymbol{x}_n)\right\}^2$$

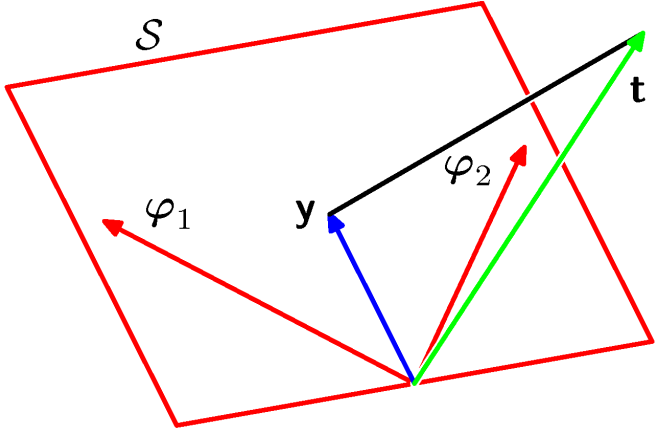
- Setting derivative wrt. w_0 to zero:

$$w_0 = \frac{1}{N}\sum_{n=1}^{N} t_n - \sum_{j=1}^{M-1} w_j \frac{1}{N}\sum_{n=1}^{N}\phi_j(\boldsymbol{x}_n) = \bar{t} - \sum_{j=1}^{M-1} w_j \overline{\phi_j}$$

- Precision β:

$$\frac{1}{\beta_{ML}} = \frac{1}{N}\sum_{n=1}^{N}\{t_n - \boldsymbol{w}_{ML}^T \boldsymbol{\phi}(\boldsymbol{x}_n)\}^2$$

Linear Basis Function Models



- *Geometry of the least squares*
- N-dimensional space spanned by axes with values $(t_1 \cdots t_N)$ defining vector $\mathbf{t}$
- Subspace defined by basis functions $\phi_j(\boldsymbol{x})$
- Represented as a vector $\boldsymbol{\phi}_j$ with Nelements $\phi_j(\boldsymbol{x}_n)$
- If $N > M$, vectors $\boldsymbol{\phi}_j$ span a linear subspace $\mathcal{S}$ of M dimensions
- $\mathbf{y}$ is N-dimensional vector with n-th element is $y(\boldsymbol{x}_n, \boldsymbol{w})$ as linear combination of ϕ_j
- Minimizing sum of squares brings $\mathbf{y}$ to minimal Euclidean distance (or its square)
- Such $\mathbf{y}$ is orthogonal projection of $\mathbf{t}$ onto subspace $\mathcal{S}$

Linear Basis Function Models

- *Sequential learning and gradient descent*
- Error function $E = \sum_n E_n$ after presentation of pattern n

$$\boldsymbol{w}^{(\tau+1)} = \boldsymbol{w}^{(\tau)} - \eta \nabla E_n$$

- η is the learning rate
- For sum of squares error *Least-Mean Squares (LMS)*

$$\boldsymbol{w}^{(\tau+1)} = \boldsymbol{w}^{(\tau)} - \eta \left(t_n - \boldsymbol{w}^{(\tau)^T} \boldsymbol{\phi}(\boldsymbol{x}_n) \right) \boldsymbol{\phi}(\boldsymbol{x}_n)$$

Linear Basis Function Models

- Regularized least squares

$$E = E_D(\boldsymbol{w}) + \lambda E_W(\boldsymbol{w})$$

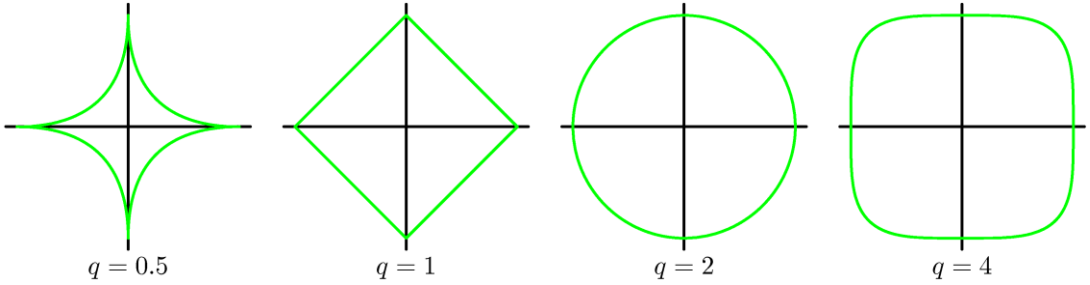
$$= \frac{1}{2}\sum_{n=1}^{N}\{t_n - \boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_n)\}^2 + \lambda\frac{1}{2}\boldsymbol{w}^T\boldsymbol{w}$$

$$\boldsymbol{w} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \lambda\boldsymbol{I})^{-1}\boldsymbol{\Phi}^T\boldsymbol{t}$$

Linear Basis Function Models

- Regularized least squares general form

$$E = \frac{1}{2}\sum_{n=1}^{N}\{t_n - \boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_n)\}^2 + \frac{\lambda}{2}\sum_{j=1}^{M}|w_j|^q$$



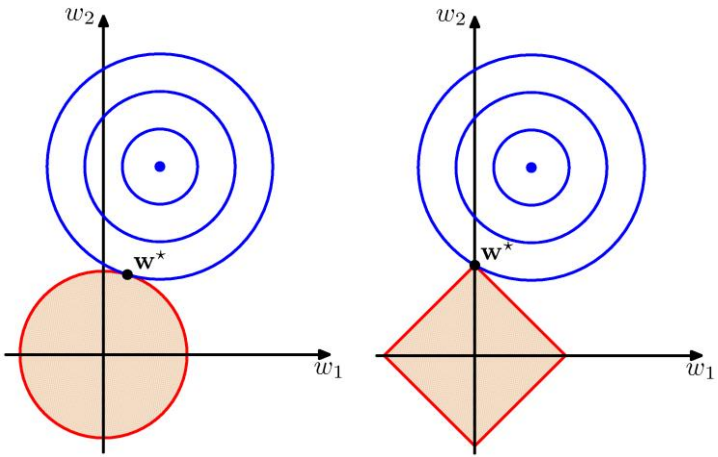
Lasso Quadratic

Linear Basis Function Models

- *Lasso Regularization* $q = 1$
- Leads to sparse $\boldsymbol{w}$ with many coefficients equal 0

$$\sum_{j=1}^{M} |w_j|^q \leq \eta$$

- Regularization is related to Lagrangian multipliers
- With increasing λ more coefficients become 0



Linear Basis Function Models

- Multiple Outputs
- Predicting $K > 1$ variables forming vector $\boldsymbol{t}$

$$\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{w}) = \boldsymbol{W}^T \boldsymbol{\phi}(\boldsymbol{x})$$

- $\boldsymbol{W}$ is an $M \times K$ matrix
- Isotropic Gaussian

$$p(\boldsymbol{t}|\boldsymbol{x}, \boldsymbol{W}, \beta) = \mathcal{N}(\boldsymbol{W}^T \boldsymbol{\phi}(\boldsymbol{x}), \beta^{-1}\boldsymbol{I})$$

Linear Basis Function Models

- Multiple Outputs
- For representing $\boldsymbol{t}_1 \dots \boldsymbol{t}_N$ observations: $\boldsymbol{T}$ is $N \times K$ matrix with n-th row ${\boldsymbol{t}_n}^T$ and $\boldsymbol{x}_1 \dots \boldsymbol{x}_N$ is matrix $\boldsymbol{X}$
- MLE and log likelihood

$$\ln p(\boldsymbol{T}|\boldsymbol{X}, \boldsymbol{W}, \beta) = \sum_{n-1}^{N} \ln \mathcal{N}(\boldsymbol{t}_n | \boldsymbol{W}^T \boldsymbol{\phi}(\boldsymbol{x}), \beta^{-1}\boldsymbol{I}) =$$

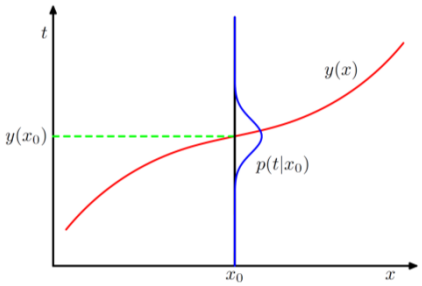
$$= \frac{NK}{2} \ln \frac{\beta}{2\pi} - \frac{\beta}{2} \sum_{n-1}^{N} \|\boldsymbol{t}_n - \boldsymbol{W}^T \boldsymbol{\phi}(\boldsymbol{x})\|^2$$

$\boldsymbol{W}_{ML} = (\boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \boldsymbol{T}$ for each target variable $\boldsymbol{t}_k$: $\boldsymbol{w}_k = (\boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \boldsymbol{t}_k$

Bias-Variance Decomposition

- Regression function (unknown, represented by data):

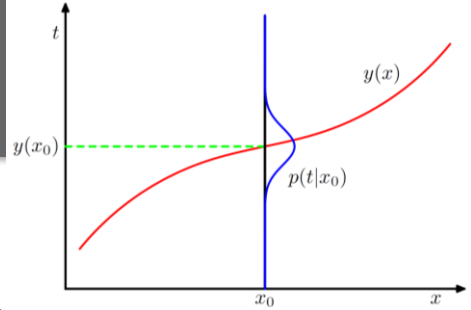
$$h(\boldsymbol{x}) = \int_t tp(t|\boldsymbol{x})dt = \mathbb{E}_t[t|\boldsymbol{x}]$$



Solution function $y(\boldsymbol{x})$:

$$(y(\boldsymbol{x}) - t)^2 = (y(\boldsymbol{x}) - h(\boldsymbol{x}) + h(\boldsymbol{x}) - t)^2$$
$$= (y(\boldsymbol{x}) - h(\boldsymbol{x}))^2 + 2(y(\boldsymbol{x}) - h(\boldsymbol{x}))(h(\boldsymbol{x}) - t)$$
$$+ (h(\boldsymbol{x}) - t)^2$$

Bias-Variance Decomposition



- Loss function:

$$\mathbb{E}(L) = \int_x \int_t (y(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x} = \int_x \int_t (y(\boldsymbol{x}) - h(\boldsymbol{x}) + h(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x} =$$

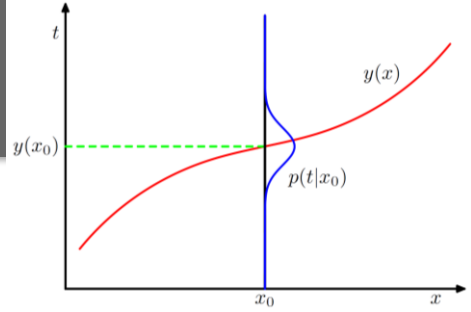
$$= \int_{\boldsymbol{x}} \big(y(\boldsymbol{x}) - h(\boldsymbol{x})\big)^2 p(\boldsymbol{x}) d\boldsymbol{x} + 2 \int_x \big(y(\boldsymbol{x}) - h(\boldsymbol{x})\big) p(\boldsymbol{x}) d\boldsymbol{x} \int_x \int_t (h(\boldsymbol{x}) \overbrace{- t}^{0}) p(t|\boldsymbol{x}) dt \, p(\boldsymbol{x}) d\boldsymbol{x}$$

$$+ \int_x \int_t (h(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x}$$

$$\mathbb{E}_t[t|\boldsymbol{x}]$$

$$\mathbb{E}(L) = \int_{\boldsymbol{x}} (y(\boldsymbol{x}) - h(\boldsymbol{x}))^2 p(\boldsymbol{x}) d\boldsymbol{x} + \int_x \int_t (h(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x}$$

Bias-Variance Decomposition



- Loss function:

$$\mathbb{E}(L)$$

$$= \int_{\boldsymbol{x}} (y(\boldsymbol{x}) - h(\boldsymbol{x}))^2 p(\boldsymbol{x}) d\boldsymbol{x} + \int_{x} \int_{t} (h(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x}$$

Choosing the optimal $y(\boldsymbol{x})$ to minimize the expected loss.

Intrinsic data noise, minimal achievable value of the expected loss.

Bias-Variance Decomposition

- Number of datasets of size N taken from $p(\boldsymbol{x}, t)$
- For each we learn $y(\boldsymbol{x}; \mathcal{D})$

$$\{y(\boldsymbol{x};\mathcal{D}) - h(\boldsymbol{x})\}^2 = \{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] + \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}^2$$

$$= \{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}^2$$

$$+2\,\{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}\{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}$$

$$+\{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}^2$$

$$\mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - h(\boldsymbol{x})\}^2]$$

$$= \{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}^2 + \mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}^2]$$

$$+\,2\mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}\underbrace{\{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}}_{=0}]$$

Bias-Variance Decomposition

$$\mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - h(\boldsymbol{x})\}^2] = \{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}^2$$
$$+\mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}^2]$$

Bias^2

Variance

- Bias represents the extent to which average prediction over all datasets differs from desired regression function
- Variance measures to which extent solutions from different datasets vary around their average

Bias-Variance Decomposition

$$\mathbb{E}[L] = \int \{\mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})] - h(\boldsymbol{x})\}^2 p(\boldsymbol{x}) d\boldsymbol{x} +$$

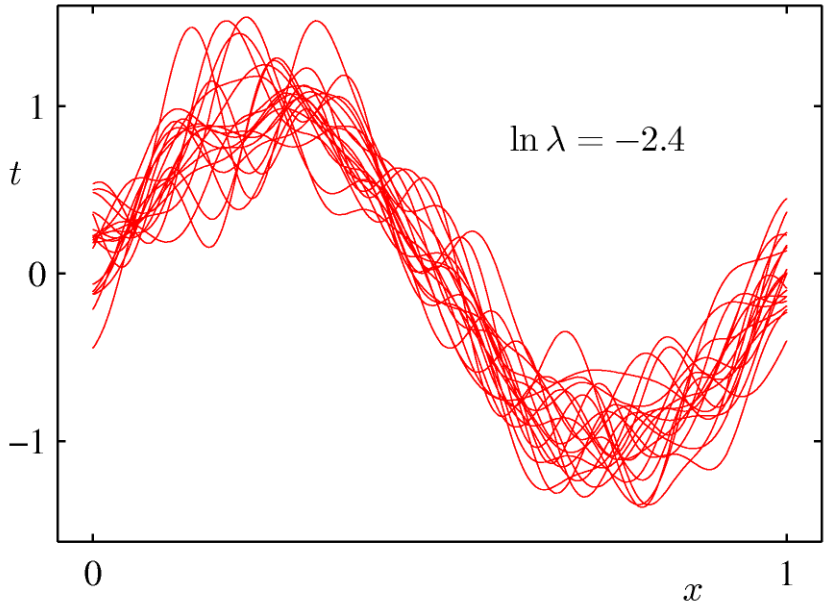
$$\int \mathbb{E}_{\mathcal{D}}[\{y(\boldsymbol{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\boldsymbol{x};\mathcal{D})]\}^2]\, p(\boldsymbol{x}) d\boldsymbol{x}$$

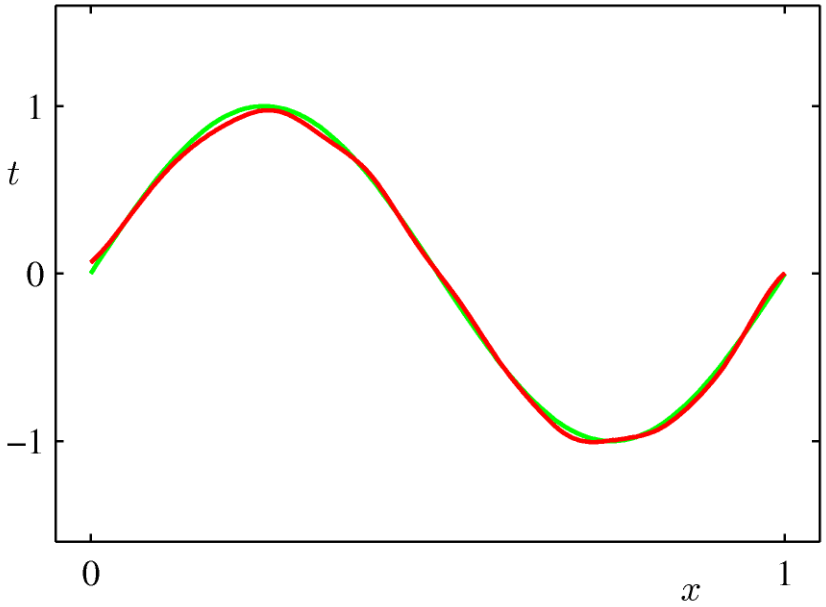
$$+ \int_{x} \int_{t} (h(\boldsymbol{x}) - t)^2 p(\boldsymbol{x}, t) dt d\boldsymbol{x}$$

Noise

Bias-Variance Decomposition

- 100 (20) datasets with $N = 25$ data points, with 24 Gaussian Basis functions





Bias-Variance Decomposition

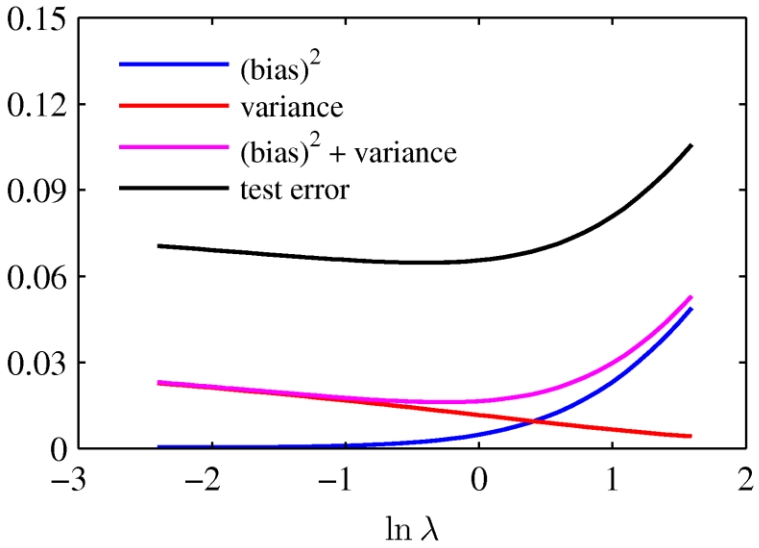
$$\bar{y}(x) = \frac{1}{L}\sum_{l=1}^{L} y^{(l)}(x)$$

Bias^2:

$$\frac{1}{N}\sum_{n=1}^{N}\{\bar{y}(x_n) - h(x_n)\}^2$$

Variance:

$$\frac{1}{N}\sum_{n=1}^{N}\frac{1}{L}\sum_{l=1}^{L}\{y^{(l)}(x_n) - \bar{y}(x_n)\}^2$$



Bayesian Linear Regression

- Likelihood function:

$$p(\mathbf{t}|\boldsymbol{\Phi},\boldsymbol{w},\beta) = \mathcal{N}(\mathbf{t}|\boldsymbol{\Phi}\boldsymbol{w},\beta^{-1}\boldsymbol{I}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \exp\left(-\frac{\beta}{2}\|(\mathbf{t}-\boldsymbol{\Phi}\boldsymbol{w}\|^2\right) =$$

$$= \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \exp\left(-\frac{\beta}{2}\sum_{n=1}^{N}(t_n - \boldsymbol{\phi}(\boldsymbol{x}_n)^T\boldsymbol{w})^2\right) = \prod_{n=1}^{N}\mathcal{N}(t_n|\boldsymbol{\phi}(\boldsymbol{x}_n)^T\boldsymbol{w},\beta^{-1})$$

- Prior:

$$p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w}|\boldsymbol{m}_0,\boldsymbol{S}_0)$$

Bayesian Linear Regression

- Conditional Gaussian (see Chapter 2):

$$p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}) \Rightarrow \boldsymbol{x} = \boldsymbol{w}, \boldsymbol{\mu} = \boldsymbol{m}_0, \boldsymbol{\Lambda}^{-1} = \boldsymbol{S}_0$$

$$p(\boldsymbol{y}|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{y}|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \boldsymbol{L}^{-1}) \Rightarrow \boldsymbol{y} = \boldsymbol{t}, \boldsymbol{A} = \boldsymbol{\Phi}, \boldsymbol{b} = \boldsymbol{0}, \boldsymbol{L} = \beta \boldsymbol{I}$$

$$p(\boldsymbol{x}|\boldsymbol{y}) = \mathcal{N}(\boldsymbol{x}|(\boldsymbol{\Lambda} + \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A})^{-1}(\boldsymbol{\Lambda}\boldsymbol{\mu} + \boldsymbol{A}^T\boldsymbol{L}(\boldsymbol{y} - \boldsymbol{b})), (\boldsymbol{\Lambda} + \boldsymbol{A}^T\boldsymbol{L}\mathbf{A})^{-1})$$

$$p(\boldsymbol{w}|\mathbf{t}) = \mathcal{N}(\boldsymbol{w}|\boldsymbol{m}_N, \boldsymbol{S}_N)$$

$$\boldsymbol{S}_N{}^{-1} = \boldsymbol{S}_0{}^{-1} + \boldsymbol{\Phi}^T\beta\boldsymbol{I}\boldsymbol{\Phi} = \boldsymbol{S}_0{}^{-1} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}$$

$$\boldsymbol{m}_N = \boldsymbol{S}_N\left(\boldsymbol{S}_0{}^{-1}\boldsymbol{m}_0 + \boldsymbol{\Phi}^T\beta\boldsymbol{I}\boldsymbol{t}\right) = \boldsymbol{S}_N\left(\boldsymbol{S}_0{}^{-1}\boldsymbol{m}_0 + \beta\boldsymbol{\Phi}^T\mathbf{t}\right)$$

Bayesian Linear Regression

- If we have infinitely broad prior $\boldsymbol{S}_0 = \alpha^{-1}\boldsymbol{I}$ where $\alpha \to 0 \Rightarrow \boldsymbol{m}_N = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{t} = \boldsymbol{w}_{ML}$
- For $N = 0 \Rightarrow \boldsymbol{m}_N = \boldsymbol{m}_0$
- For prior $p(\boldsymbol{w}|\alpha) = \mathcal{N}(\boldsymbol{w}|\mathbf{0}, \alpha^{-1}\boldsymbol{I})$:

$$\boldsymbol{S}_N{}^{-1} = \alpha\boldsymbol{I} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}$$

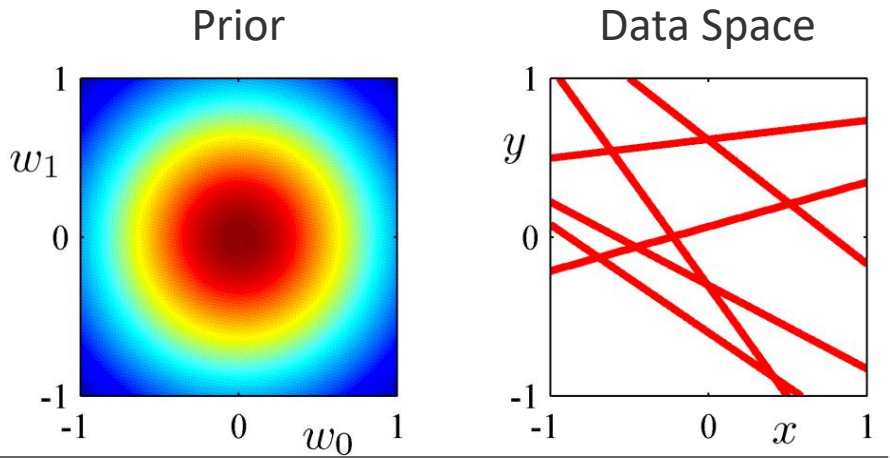
$$\boldsymbol{m}_N = \beta\boldsymbol{S}_N\boldsymbol{\Phi}^T\mathbf{t}$$

$$\ln p(\boldsymbol{w}|\mathbf{t}) = -\frac{\beta}{2}\sum_{n=1}^{N}\{t_n - \boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_n)\}^2 - \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w} + c$$

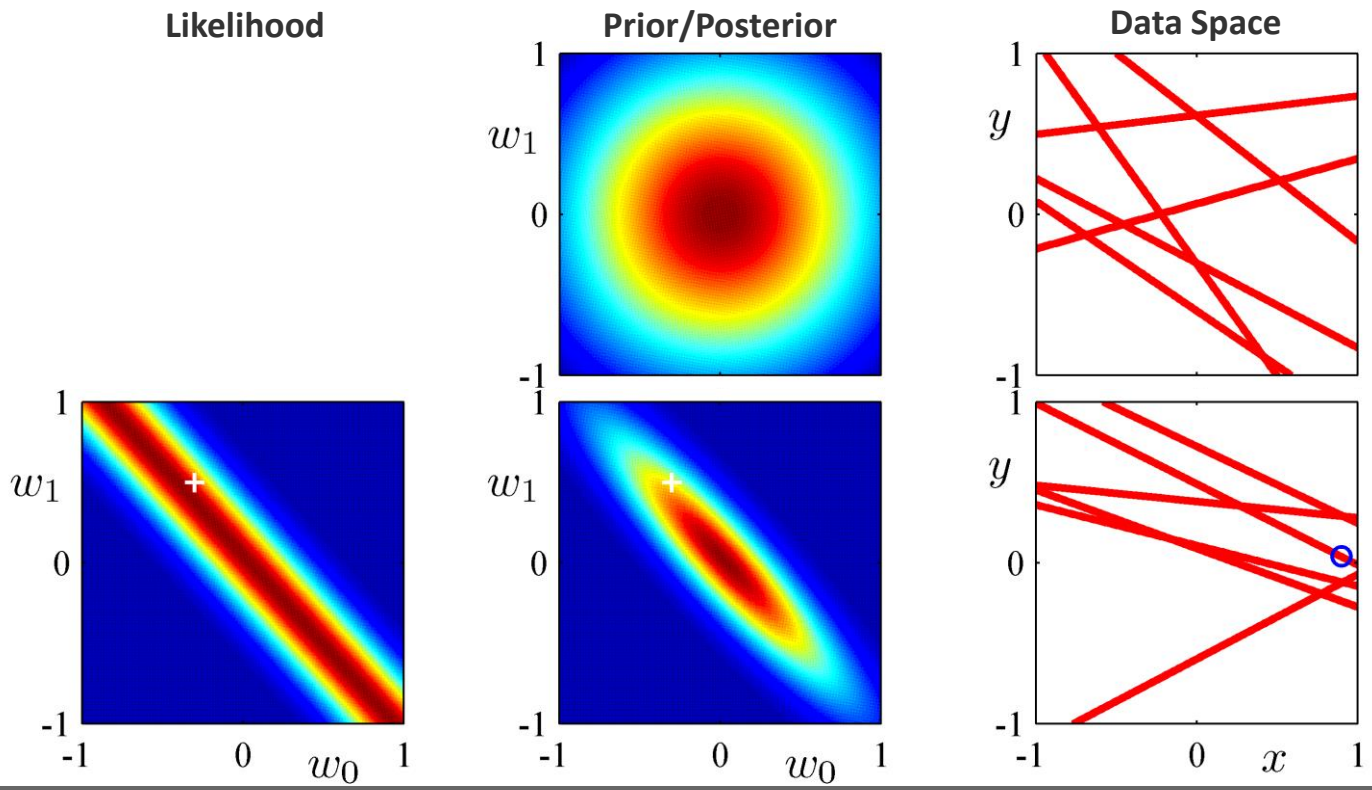
- Corresponds to sum of squares error using $\lambda = \frac{\alpha}{\beta}$

Bayesian Linear Regression

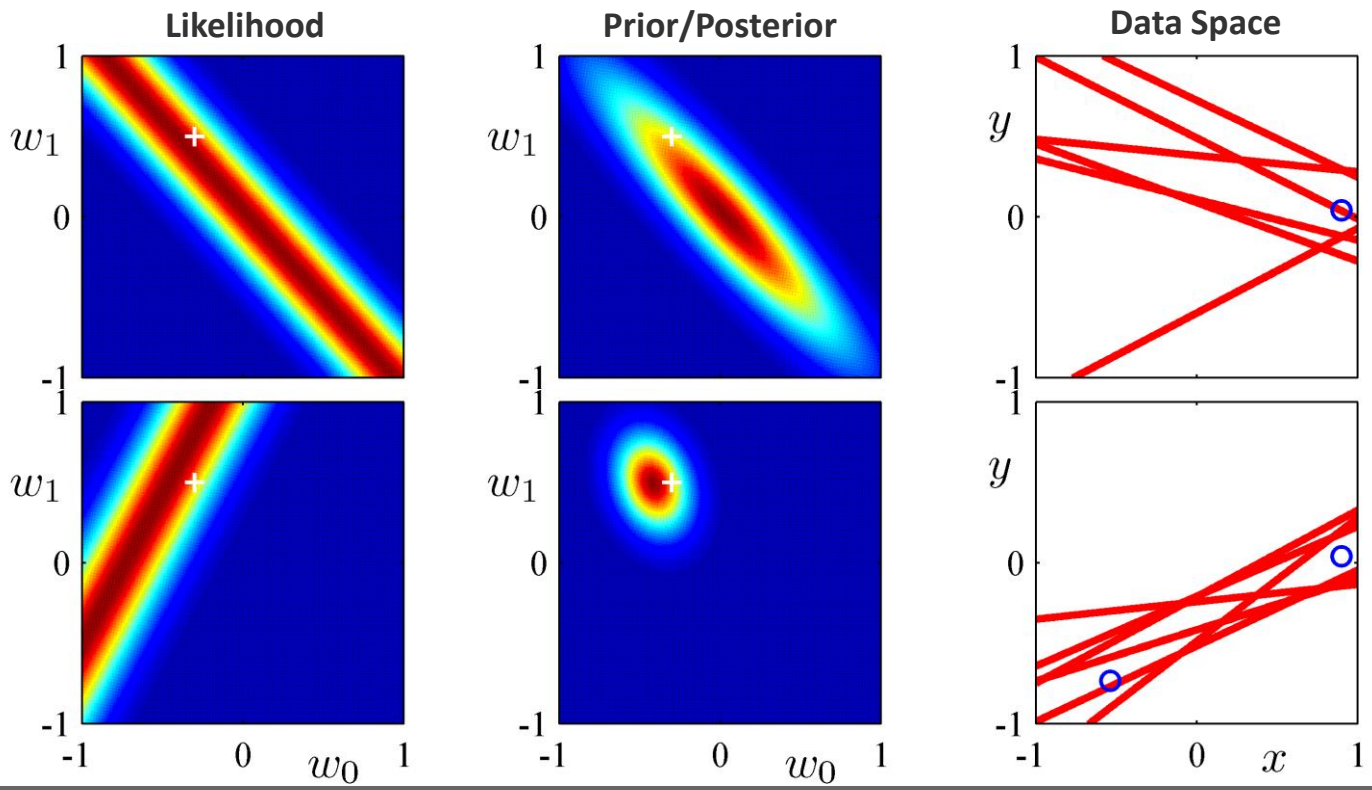
- Linear regression $y(x, \boldsymbol{w}) = w_0 + w_1 x$ from samples of function $f(x, \boldsymbol{a}) = -0.3 + 0.5x$ and choosing $x_n \in U(x| -1,1)$ and adding noise $\beta = 25$ and for prior $\alpha = 2$



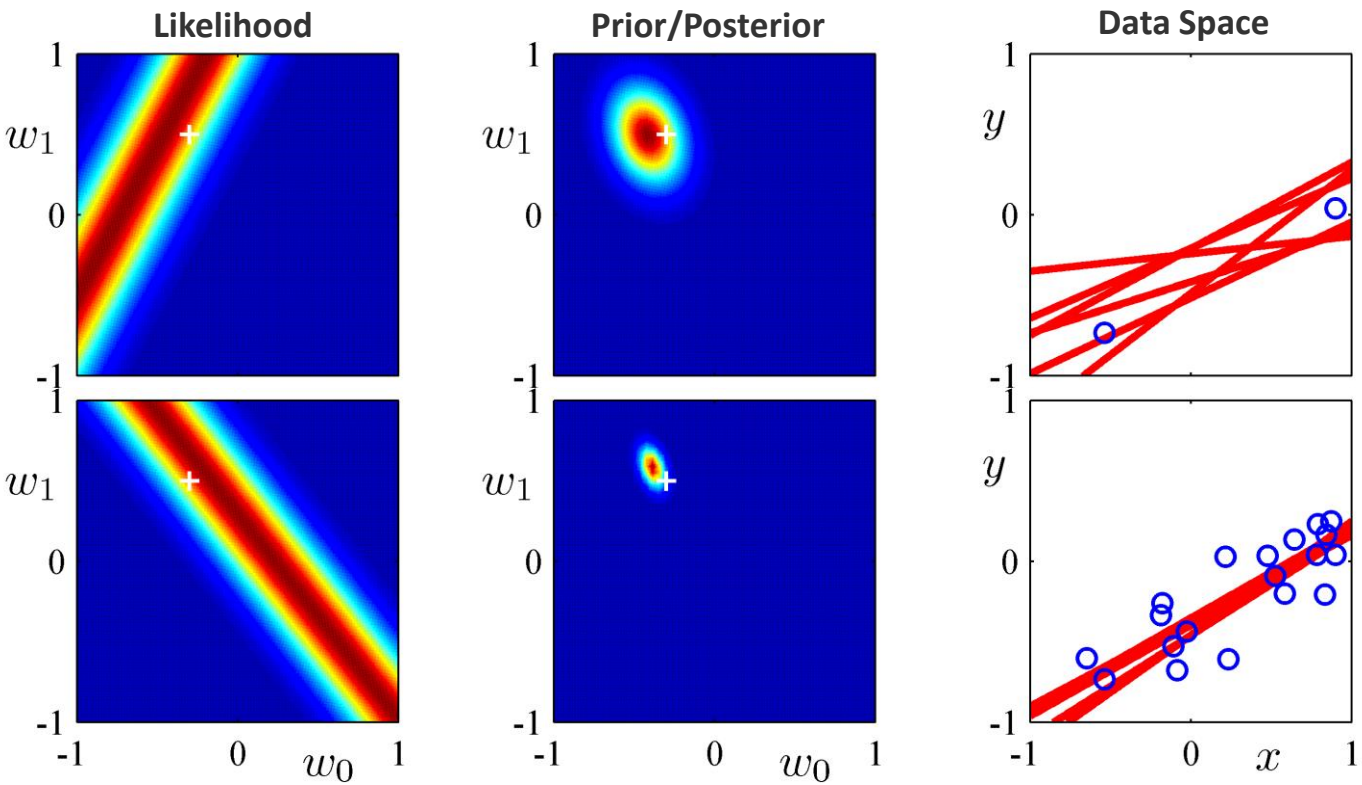
Bayesian Linear Regression



Bayesian Linear Regression



Bayesian Linear Regression



Predictive Distribution

- Integral over conditional distribution of the target variable and posterior weight distribution

$$p(t|\boldsymbol{x}, \mathbf{X}, \mathbf{t}, \alpha, \beta) = \int p(t|\boldsymbol{x}, \boldsymbol{w}, \beta) p(\boldsymbol{w}|\mathbf{X}, \mathbf{t}, \alpha, \beta) d\boldsymbol{w} = \int \mathcal{N}(t|\boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}), \beta^{-1}) \mathcal{N}(\boldsymbol{w}|\boldsymbol{m}_N, \boldsymbol{S}_N) d\boldsymbol{w}$$

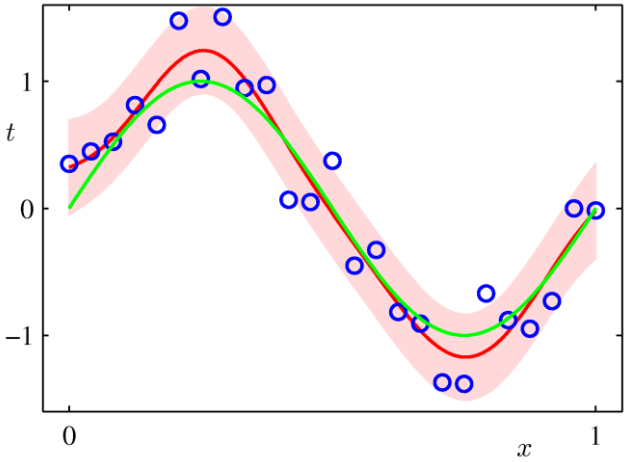
$$= \mathcal{N}\left(t \middle| \boldsymbol{m}_N{}^T \boldsymbol{\phi}(\boldsymbol{x}), \frac{1}{\beta} + \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\phi}(\boldsymbol{x})\right)$$

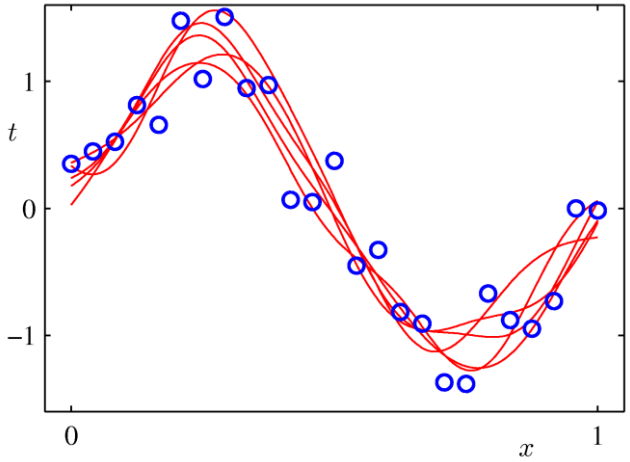
- If we search for mean of a Linear Gaussian for $\boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})$, it is equal to $\boldsymbol{m}_N{}^T \boldsymbol{\phi}(\boldsymbol{x})$
- Variance contains noise term and uncertainty associated with parameters $\boldsymbol{w}$. As these two distributions are independent variances are additive
- Remember: $\mathrm{cov}[\boldsymbol{y}] = \boldsymbol{L}^{-1} + \boldsymbol{A}\boldsymbol{\Lambda}^{-1}\boldsymbol{A}^T$: $\mathrm{var}\left(\epsilon + \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})\right) = \mathrm{var}(\epsilon) + \mathrm{var}\left(\boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})\right) = \beta^{-1} + \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\phi}(\boldsymbol{x})$

Predictive Distribution

- Example: Sinusoidal data, 9 Gaussian basis functions

$$p(t|x, \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}\left(t \middle| \boldsymbol{m}_N{}^T \boldsymbol{\phi}(\boldsymbol{x}), \frac{1}{\beta} + \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\phi}(\boldsymbol{x})\right)$$





Left: mean and variance as a function of x given $N = 1, 2, 4, 25$. Right: plots of $y(x, \boldsymbol{w})$ sampled from posterior over $\boldsymbol{w}$.

Equivalent Kernel

- For prior $p(\boldsymbol{w}|\alpha) = \mathcal{N}(\boldsymbol{w}|\boldsymbol{0}, \alpha^{-1}\boldsymbol{I})$: $\boldsymbol{m}_N = \beta \boldsymbol{S}_N \boldsymbol{\Phi}^T \mathbf{t}$ posterior mean
- Precision $\boldsymbol{S}_N{}^{-1} = \boldsymbol{S}_0{}^{-1} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi}$
- For Linear regression

$$y(\boldsymbol{x}, \boldsymbol{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\boldsymbol{x}) = \sum_{j=0}^{M-1} w_j \phi_j(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})$$

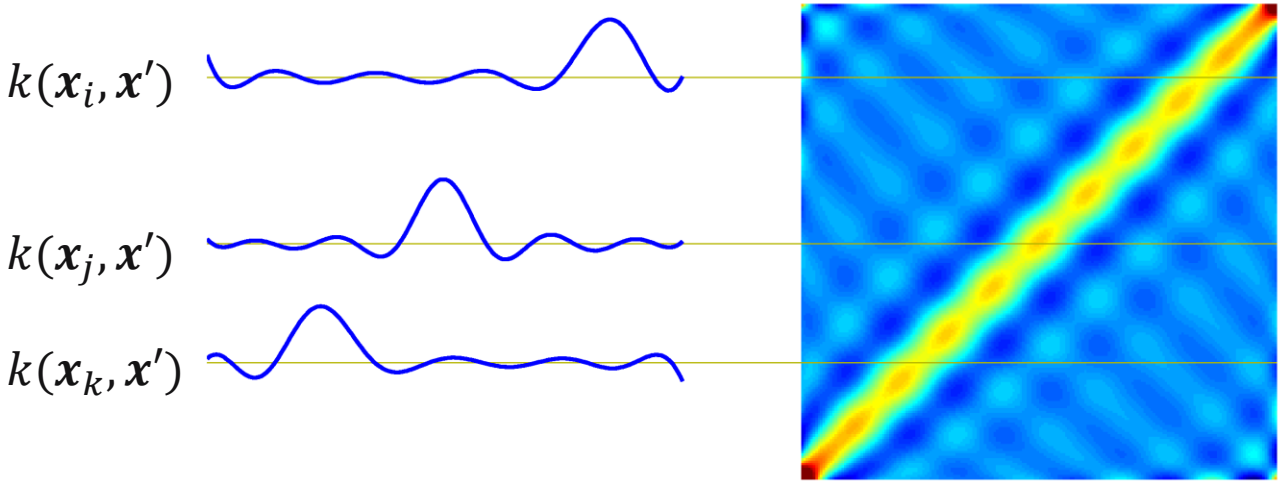
For mean weights:

$$y(\boldsymbol{x}, \boldsymbol{m}_N) = \boldsymbol{m}_N{}^T \boldsymbol{\phi}(\boldsymbol{x}) = \beta \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\Phi}^T \mathbf{t} = \sum_{n=1}^{N} \beta \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\phi}(\boldsymbol{x}_n) t_n = \sum_{n=1}^{N} k(\boldsymbol{x}, \boldsymbol{x}_n) t_n$$

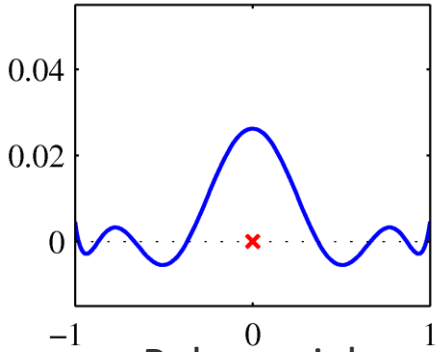
where $k(\boldsymbol{x}, \boldsymbol{x}') = \beta \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{S}_N \boldsymbol{\phi}(\boldsymbol{x}')$

Equivalent Kernel

- Weight of t_n depends on $\|\boldsymbol{x} - \boldsymbol{x}'\|$
- Is a function of $\boldsymbol{x}'$

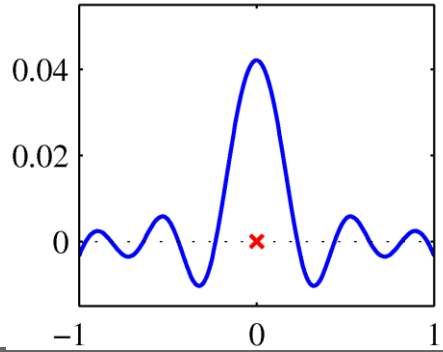


Gaussian



Polynomial

Sigmoidal

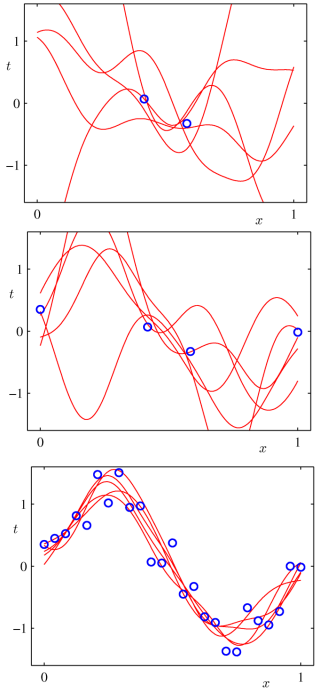


Equivalent Kernel

- Predictive covariance of outputs for any $\boldsymbol{x}$ and $\boldsymbol{x}'$:

$$\mathrm{cov}[y(\boldsymbol{x}), y(\boldsymbol{x}')] = \mathrm{cov}[\boldsymbol{\phi}(\boldsymbol{x})^T\boldsymbol{w}, \boldsymbol{\phi}(\boldsymbol{x}')^T\boldsymbol{w}]$$
$$= \mathrm{cov}[\boldsymbol{\phi}(\boldsymbol{x})^T\boldsymbol{w}, \boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}')] = \mathbb{E}[\boldsymbol{\phi}(\boldsymbol{x})^T\boldsymbol{w}\boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}')]$$
$$= \boldsymbol{\phi}(\boldsymbol{x})^T\mathbb{E}[\boldsymbol{w}\boldsymbol{w}^T]\boldsymbol{\phi}(\boldsymbol{x}') = \boldsymbol{\phi}(\boldsymbol{x})^T\boldsymbol{S}_N\boldsymbol{\phi}(\boldsymbol{x}') = \beta^{-1}k(\boldsymbol{x}, \boldsymbol{x}')$$

- Drawing samples from posterior over $\boldsymbol{w}$ and plotting $y(\boldsymbol{x}, \boldsymbol{w})$ shows joint uncertainty in posterior between y values at 2 or more x values as governed by equivalent kernel
- Instead of introducing weighted basis functions, we define a localized kernel directly for predicting output values for new input $\boldsymbol{x}$.



Equivalent Kernel

- The weights of the kernel sum to 1:

$$\sum_{n=1}^{N} k(\boldsymbol{x}, \boldsymbol{x}_n) = 1$$

- If all $t_n = 1$ then the bias term $\phi_0(\boldsymbol{x}) = 1$ the function $y(\boldsymbol{x}) = 1$ can be perfectly represented. Predictive mean $\hat{y}(\boldsymbol{x}) = 1$:

$$\hat{y}(\boldsymbol{x}) = \sum_{n=1}^{N} k(\boldsymbol{x}, \boldsymbol{x}_n) t_n = \sum_{n=1}^{N} k(\boldsymbol{x}, \boldsymbol{x}_n) = 1$$

Equivalent kernel can be written as $k(\boldsymbol{x}, \boldsymbol{x}') = \boldsymbol{\psi}(\boldsymbol{x})^T \boldsymbol{\psi}(\boldsymbol{x}')$ where

$$\boldsymbol{\psi}(\boldsymbol{x}) = \sqrt{\beta \boldsymbol{S}_N} \boldsymbol{\phi}(\boldsymbol{x})$$

Bayesian Model Comparison

- Allowing comparison of models without the validation set so all data can be used for training
- Set of $\{\mathcal{M}_i\}$ models $i = 1..L$ where $\mathcal{M}_i$ expresses the probability distribution defined over the target $\mathbf{t}$ with known $\boldsymbol{X}$ (or over joint probability $\boldsymbol{X}$,$\mathbf{t}$).
- A prior $p(\mathcal{M}_i)$ defines a (uniform) preference of a model

Bayesian Model Comparison

- Evaluating posterior distribution allows for predictive distribution:

$$p(\mathcal{M}_i|\mathcal{D}) \propto p(\mathcal{D}|\mathcal{M}_i)p(\mathcal{M}_i)$$

$$p(t|\boldsymbol{x},\mathcal{D}) = \sum_{i=1}^{L} p(t|\boldsymbol{x},\mathcal{M}_i,\mathcal{D})p(\mathcal{M}_i|\mathcal{D})$$

- Obtained as averaging predictive distributions of individual models weighted by posterior probabilities $p(\mathcal{M}_i|\mathcal{D})$
- *Model Selection*: approximation to averaging is to pick most likely model
- *Bayes factor*: $BF(\mathcal{D}) = p(\mathcal{D}|\mathcal{M}_i)/p(\mathcal{D}|\mathcal{M}_j)$

Bayesian Model Comparison

- The concept $p(\mathcal{D}|\mathcal{M}_i)$ is *model evidence* which expresses how well the model $\mathcal{M}_i$ (including its prior assumptions) explains the entire dataset $\mathcal{D}$, without committing to any single choice of parameters $\boldsymbol{w}$.
- Also known as marginal likelihood: a likelihood function over a space of models where parameters have been marginalized out:

$$p(\mathcal{D}|\mathcal{M}_i) = \int p(\mathcal{D}|\boldsymbol{w}, \mathcal{M}_i) p(\boldsymbol{w}|\mathcal{M}_i) d\boldsymbol{w}$$

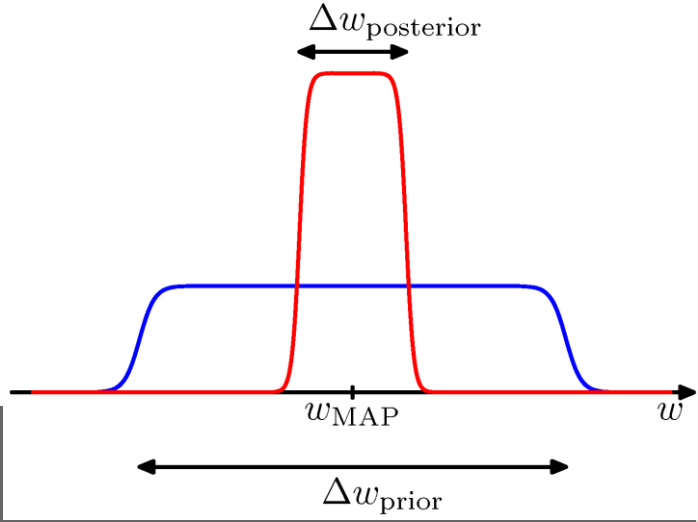
- Model evidence is also the normalizing term in evaluating posterior distribution:

$$p(\boldsymbol{w}|\mathcal{D}, \mathcal{M}_i) = \frac{p(\mathcal{D}|\boldsymbol{w}, \mathcal{M}_i) p(\boldsymbol{w}|\mathcal{M}_i)}{p(\mathcal{D}|\mathcal{M}_i)}$$

Bayesian Model Comparison

- Considering just one parameter of a model w (e.g. μ in Bernoulli)
- Observing the width $\Delta w_{posterior}$ of the peak around w_{MAP} for posterior proportional to $p(\mathcal{D}|w)p(w)$.
- Prior is flat with width Δw_{prior}

$$p(\mathcal{D}) = \int p(\mathcal{D}|w)p(w)dw = \frac{1}{\Delta w_{prior}} \int p(\mathcal{D}|w)dw \simeq p(\mathcal{D}|w_{MAP}) \frac{\Delta w_{posterior}}{\Delta w_{prior}}$$



Bayesian Model Comparison

- Taking a log:

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|w_{MAP}) + \ln \frac{\Delta w_{posterior}}{\Delta w_{prior}}$$

- First term stands for fitting data by most probable parameters (log likelihood)
- Second term is negative, penalizes model according to its complexity
- For M parameters with the same ratio:

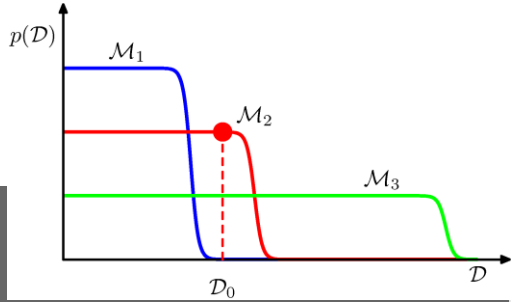
$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|w_{MAP}) + M \ln \frac{\Delta w_{posterior}}{\Delta w_{prior}}$$

- With increasing model complexity, first term decreases and the second term increases (abs val). Model evidence: trade-off between two terms

Bayesian Model Comparison

- Favoring models of intermediate complexity
- All possible datasets $\mathcal{D}$ with associated probability that $\mathcal{D}$ was generated by $\mathcal{M}_i$
- Simple model $\mathcal{M}_1$ is not capable of generating data
- Complex model $\mathcal{M}_3$ can generate many datasets so probability that it generated $\mathcal{D}_0$ is lower than with $\mathcal{M}_2$
- If the $\mathcal{D}$ is generated by $\mathcal{M}_a$ even if for fixed $\mathcal{D}$ its $BF(\mathcal{D})$ is bigger for $\mathcal{M}_b$ than for $\mathcal{M}_a$ on average we obtain expectation

$$\mathbb{E}[BF(\mathcal{D})] = \int p(\mathcal{D}|\mathcal{M}_a) \ln \frac{p(\mathcal{D}|\mathcal{M}_a)}{p(\mathcal{D}|\mathcal{M}_b)} d\mathcal{D}$$



Evidence Approximation

- Introducing hyperpriors over α and β

$$p(t|\boldsymbol{x}, \mathbf{X}, \mathbf{t}, \alpha, \beta)$$

$$= \int\int\int p(t|\boldsymbol{x}, \boldsymbol{w}, \beta)p(\boldsymbol{w}|\mathbf{X}, \mathbf{t}, \alpha, \beta)p(\alpha, \beta|\mathbf{X}, \mathbf{t})d\boldsymbol{w}d\alpha d\beta \simeq \int p\left(t\middle|\boldsymbol{x}, \boldsymbol{w}, \hat{\beta}\right)p\left(\boldsymbol{w}\middle|\mathbf{X}, \mathbf{t}, \hat{\alpha}, \hat{\beta}\right)d\boldsymbol{w}$$

$$p(\alpha, \beta|\mathbf{X}, \mathbf{t}) \propto p(\mathbf{t}|\mathbf{X}, \alpha, \beta)p(\alpha, \beta)$$

- Marginal likelihood (or evidence) function

$$p(\mathbf{t}|\mathbf{X}, \alpha, \beta) = \int p(\mathbf{t}|\mathbf{X}, \boldsymbol{w}, \beta)p(\boldsymbol{w}|\alpha)d\boldsymbol{w} = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}}\left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}}\int \exp\{-E(\boldsymbol{w})\}\,d\boldsymbol{w}$$

$$E(\boldsymbol{w}) = \beta E_D(\boldsymbol{w}) + \alpha E_w(\boldsymbol{w}) = \frac{\beta}{2}\|(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{w}\|^2 + \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w}$$

Evidence Approximation

- Vector $\boldsymbol{m}_N$ minimizes $E(\boldsymbol{w})$

$$\nabla_{\boldsymbol{w}} E(\boldsymbol{w} = \boldsymbol{m}_N) = \boldsymbol{0}$$

$$-\beta \boldsymbol{\Phi}^T(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N) + \alpha \boldsymbol{m}_N = \boldsymbol{0} \Rightarrow (\alpha \boldsymbol{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi})\boldsymbol{m}_N = \beta \boldsymbol{\Phi}^T \mathbf{t} = \boldsymbol{A}\boldsymbol{m}_N$$

$$\boldsymbol{w} = \boldsymbol{m}_N + \boldsymbol{\delta} \Rightarrow$$

$$\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{w}\|^2 = \|\mathbf{t} - \boldsymbol{\Phi}(\boldsymbol{m}_N + \boldsymbol{\delta})\|^2 = \|(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N) - \boldsymbol{\Phi}\boldsymbol{\delta}\|^2$$

$$\boldsymbol{w}^T\boldsymbol{w} = (\boldsymbol{m}_N + \boldsymbol{\delta})^T(\boldsymbol{m}_N + \boldsymbol{\delta}) = \boldsymbol{m}_N{}^T\boldsymbol{m}_N + 2\boldsymbol{m}_N{}^T\boldsymbol{\delta} + \boldsymbol{\delta}^T\boldsymbol{\delta}$$

$$E(\boldsymbol{m}_N + \boldsymbol{\delta}) = \frac{\beta}{2}\|(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N) - \boldsymbol{\Phi}\boldsymbol{\delta}\|^2 + \frac{\alpha}{2}(\boldsymbol{m}_N + \boldsymbol{\delta})^T(\boldsymbol{m}_N + \boldsymbol{\delta}) =$$

$$\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N\|^2 + \frac{\alpha}{2}\boldsymbol{m}_N{}^T\boldsymbol{m}_N + (-\beta(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N)^T\boldsymbol{\Phi}\boldsymbol{\delta}) + \alpha\boldsymbol{m}_N{}^T\boldsymbol{\delta} + \frac{\beta}{2}\boldsymbol{\delta}^T\boldsymbol{\Phi}^T\boldsymbol{\Phi}\boldsymbol{\delta} + \frac{\alpha}{2}\boldsymbol{\delta}^T\boldsymbol{\delta} =$$

$$E(\boldsymbol{m}_N) + [-\beta(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N)^T\boldsymbol{\Phi} + \alpha\boldsymbol{m}_N{}^T]\boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^T(\beta\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \alpha\boldsymbol{I})\boldsymbol{\delta} =$$

$$E(\boldsymbol{m}_N) + \frac{1}{2}(\boldsymbol{w} - \boldsymbol{m}_N)^T\boldsymbol{A}(\boldsymbol{w} - \boldsymbol{m}_N) + [-\beta(\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N)^T\boldsymbol{\Phi} + \alpha\boldsymbol{m}_N{}^T]\boldsymbol{\delta} = E(\boldsymbol{m}_N) + \frac{1}{2}(\boldsymbol{w} - \boldsymbol{m}_N)^T\boldsymbol{A}(\boldsymbol{w} - \boldsymbol{m}_N)$$

Evidence Approximation

$$\beta \boldsymbol{\Phi}^T \mathbf{t} = \boldsymbol{A}\boldsymbol{m}_N \Rightarrow \boldsymbol{m}_N = \beta \boldsymbol{A}^{-1} \boldsymbol{\Phi}^T \mathbf{t} \Rightarrow \boldsymbol{A} = \boldsymbol{S}_N{}^{-1}$$

$$\int \exp\{-E(\boldsymbol{w})\}\, d\boldsymbol{w} = \exp\{-E(\boldsymbol{m}_N)\} + \int \exp\{-\frac{1}{2}(\boldsymbol{w} - \boldsymbol{m}_N)^T \boldsymbol{A}(\boldsymbol{w} - \boldsymbol{m}_N)\}\, d\boldsymbol{w}$$

$$= \exp\{-E(\boldsymbol{m}_N)\}\, (2\pi)^{\frac{M}{2}} |\boldsymbol{A}|^{-\frac{1}{2}}$$

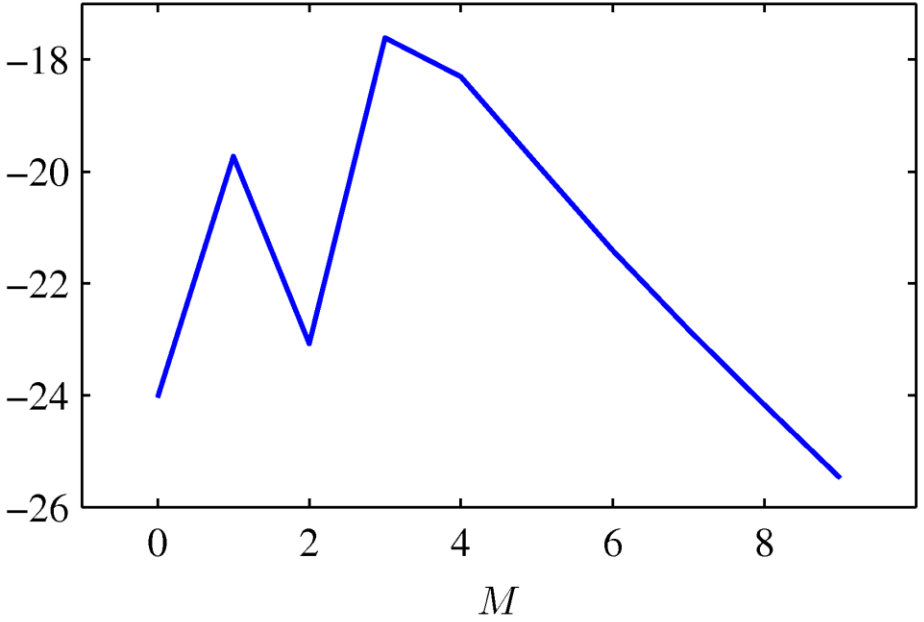
$$p(\mathbf{t}|\mathbf{X}, \alpha, \beta) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \exp\{-E(\boldsymbol{m}_N)\}\ (2\pi)^{\frac{M}{2}} |\boldsymbol{A}|^{-\frac{1}{2}}$$

$$\ln p(\mathbf{t}|\mathbf{X}, \alpha, \beta) = \frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - E(\boldsymbol{m}_N) - \frac{1}{2} \ln|\boldsymbol{A}| - \frac{N}{2} \ln 2\pi$$

Evidence Approximation

- Synthetic sinusoidal example, M-th degree polynomial
- Prior $\alpha = 0.005$
- $M = 0$ bad fit
- $M = 3$ good fit low M
- $M = 8$ good fit high M

$$\frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - E(\boldsymbol{m}_N) - \frac{1}{2}\ln|\boldsymbol{A}| - \frac{N}{2}\ln 2\pi$$



Evidence Approximation

- Maximizing Evidence Function: α

$$\boldsymbol{A} = \boldsymbol{S}_N^{-1} = \alpha \boldsymbol{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi} : \boldsymbol{\Phi}^T \boldsymbol{\Phi} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^T$$

$$\alpha \boldsymbol{I} + \beta \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^T = \mathbf{U}(\alpha \boldsymbol{I} + \beta \boldsymbol{\Lambda}) \mathbf{U}^T$$

- Diagonal entries of posterior precision: $\alpha + \beta \lambda_i$

$$\frac{d}{d\alpha} \ln |\boldsymbol{A}| = \frac{d}{d\alpha} \ln \prod_{i=1}^{M} (\alpha + \beta \lambda_i) = \sum_{i=1}^{M} \frac{1}{\alpha + \beta \lambda_i}$$

$$\frac{d}{d\alpha} \left(\frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - E(\boldsymbol{m}_N) - \frac{1}{2} \ln |\boldsymbol{A}| - \frac{N}{2} \ln 2\pi \right) = 0$$

$$= \frac{M}{2\alpha} - \frac{1}{2} \boldsymbol{m}_N{}^T \boldsymbol{m}_N - \frac{1}{2} \sum_{i=1}^{M} \frac{1}{\alpha + \beta \lambda_i}$$

- We neglect derivatives $\boldsymbol{m}_N{}^T \boldsymbol{m}_N$ wrt. α as the resulting formula is correct at stationary point

Evidence Approximation

- Maximizing Evidence Function:α

$$\alpha \boldsymbol{m}_N{}^T \boldsymbol{m}_N = M - \alpha \sum_{i=1}^{M} \frac{1}{\alpha + \beta\lambda_i} = \sum_{i=1}^{M} \frac{\beta\lambda_i}{\alpha + \beta\lambda_i} = \gamma$$

$$\alpha = \frac{\gamma}{\boldsymbol{m}_N{}^T \boldsymbol{m}_N}$$

- Parameter α is found iteratively, precomputing λ_i
- It has been found by looking at training data only

Evidence Approximation

- Maximizing Evidence Function: β

$$\frac{d}{d\beta}\ln|\boldsymbol{A}| = \frac{d}{d\beta}\ln\prod_{i=1}^{M}(\alpha+\beta\lambda_i) = \sum_{i=1}^{M}\frac{\lambda_i}{\alpha+\beta\lambda_i}$$

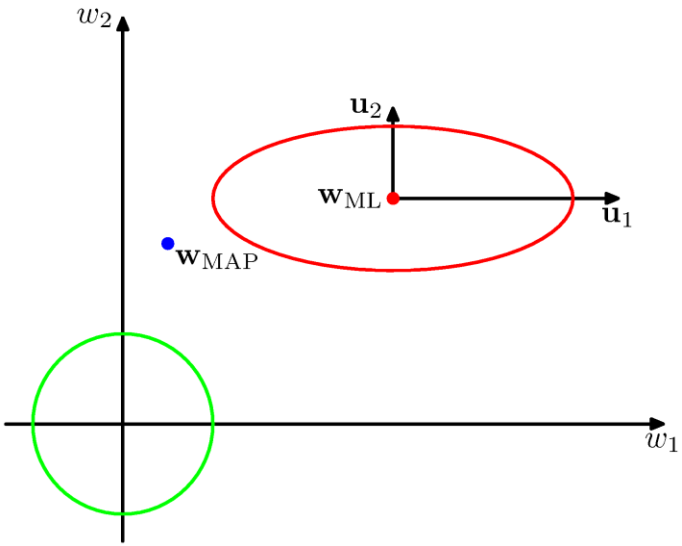
$$\frac{d}{d\beta}\left(\frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - E(\boldsymbol{m}_N) - \frac{1}{2}\ln|\boldsymbol{A}| - \frac{N}{2}\ln 2\pi\right) = 0$$

$$= \frac{N}{2\beta} - \frac{1}{2}\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N\|^2 - \frac{1}{2}\sum_{i=1}^{M}\frac{\lambda_i}{\alpha+\beta\lambda_i}$$

$$N - \beta\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N\|^2 - \gamma = 0 \Rightarrow \beta = \frac{N-\gamma}{\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{m}_N\|^2}$$

Evidence Approximation

- Positive definite matrix $\mathbf{\Phi}^T\mathbf{\Phi}$ has positive λ_i
- Small λ_i results in low curvature ~ elongated axis
- As $\frac{\beta\lambda_i}{\alpha+\beta\lambda_i} \in [0,1], \gamma \in [0, M]$
- For $\alpha \ll \beta\lambda_i \Rightarrow w_i \to w_{iML}$
- For $\alpha \gg \beta\lambda_i \Rightarrow w_i \to 0$
- Effective total number of well-determined parameters: γ

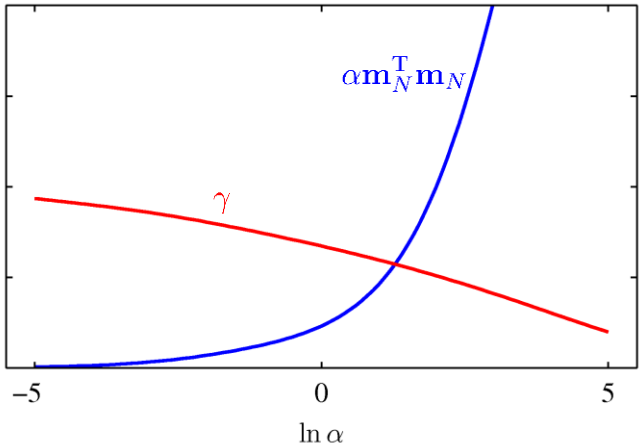


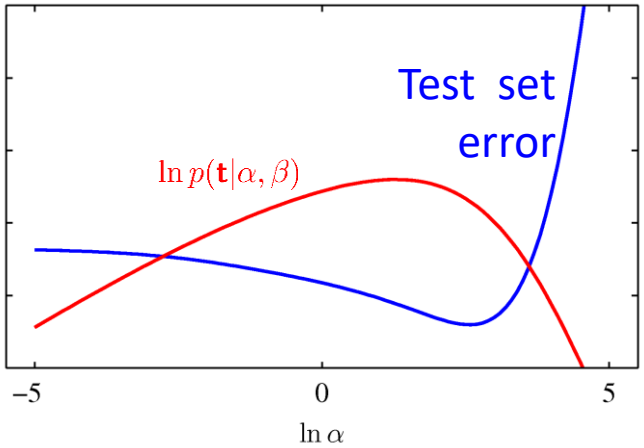
Evidence Approximation

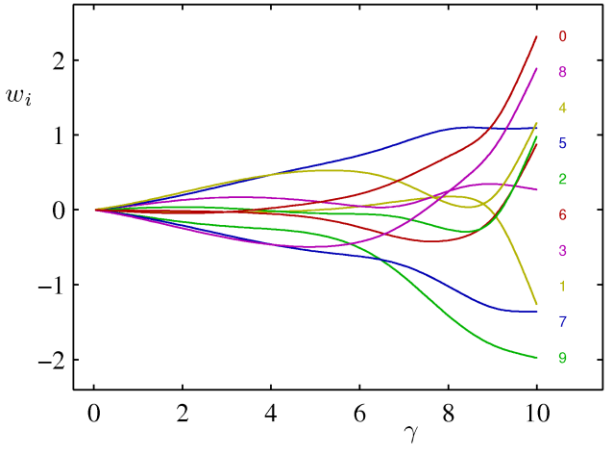
- Effective total number of well-determined parameters: γ and $M - \gamma$ are set to small value by prior
- If $\mathrm{N} \gg M$ this leads to high λ_i and $\gamma \to M$
  - $\beta = \frac{N}{\|\mathbf{t} - \boldsymbol{\Phi} \boldsymbol{m}_N\|^2}$
  - $\alpha = \frac{M}{\boldsymbol{m}_N{}^T \boldsymbol{m}_N}$

Evidence Approximation

- Synthetic sinusoidal data, 9 Gaussian basis functions, $\beta = 11.1$



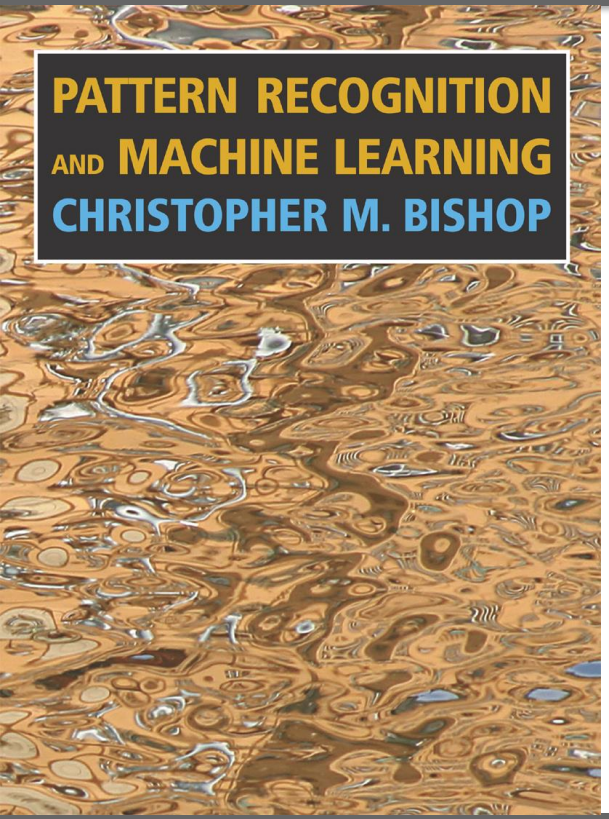




Limitations of Fixed Basis Functions

- Fixed ϕ_i is limiting as ideally basis functions are also trained
- Large amount of parameters, increasing exponentially with dimensions
- Target variables might have significant dependence only on few basis functions

Credits



Pattern Recognition and Machine Learning

by Christopher M. Bishop

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